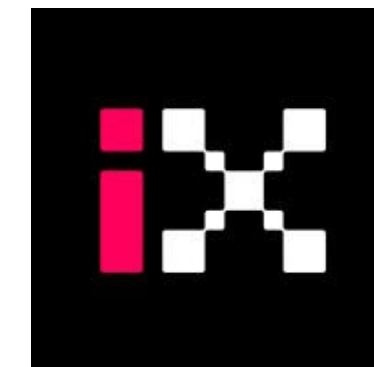
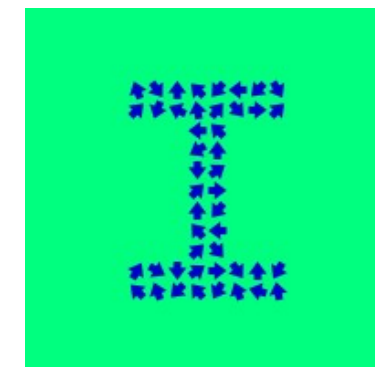


Emulating the REACH dipole beam pattern

Harry Bevins

With John Cumner, Dominic Anstey, Jean Cavillot, Jacob Tutt

IMPERIAL



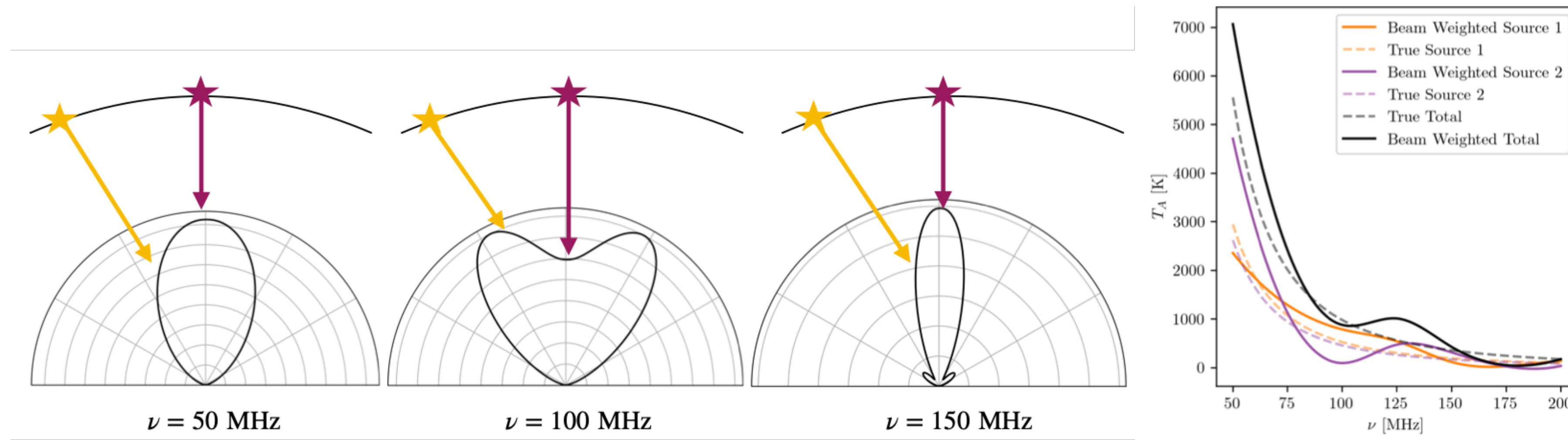
The problem...

The problem...

- Measured antenna temperature is a convolution of the beam and radio sky

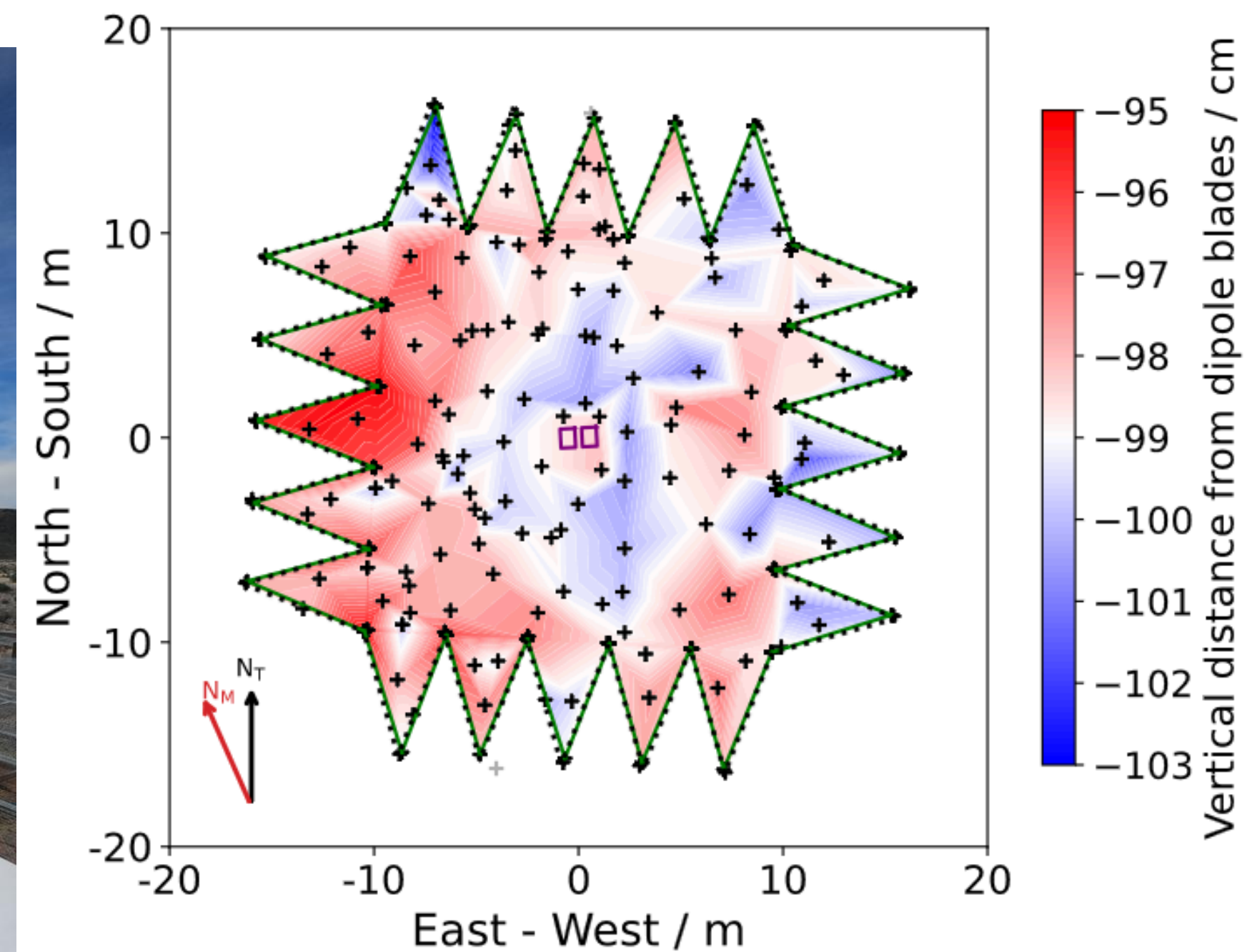
$$T_{\text{ant}}(\nu) = \int T_{\text{sky}}(\nu, \Omega) D(\nu, \Omega) d\Omega$$

- Getting the beam wrong can prevent a detection or lead to a false detection e.g. Cumner et al. 2023 and Pattison et al. 2026



The problem...

- Typically assume an idealised pattern but in practice things are a lot more complicated

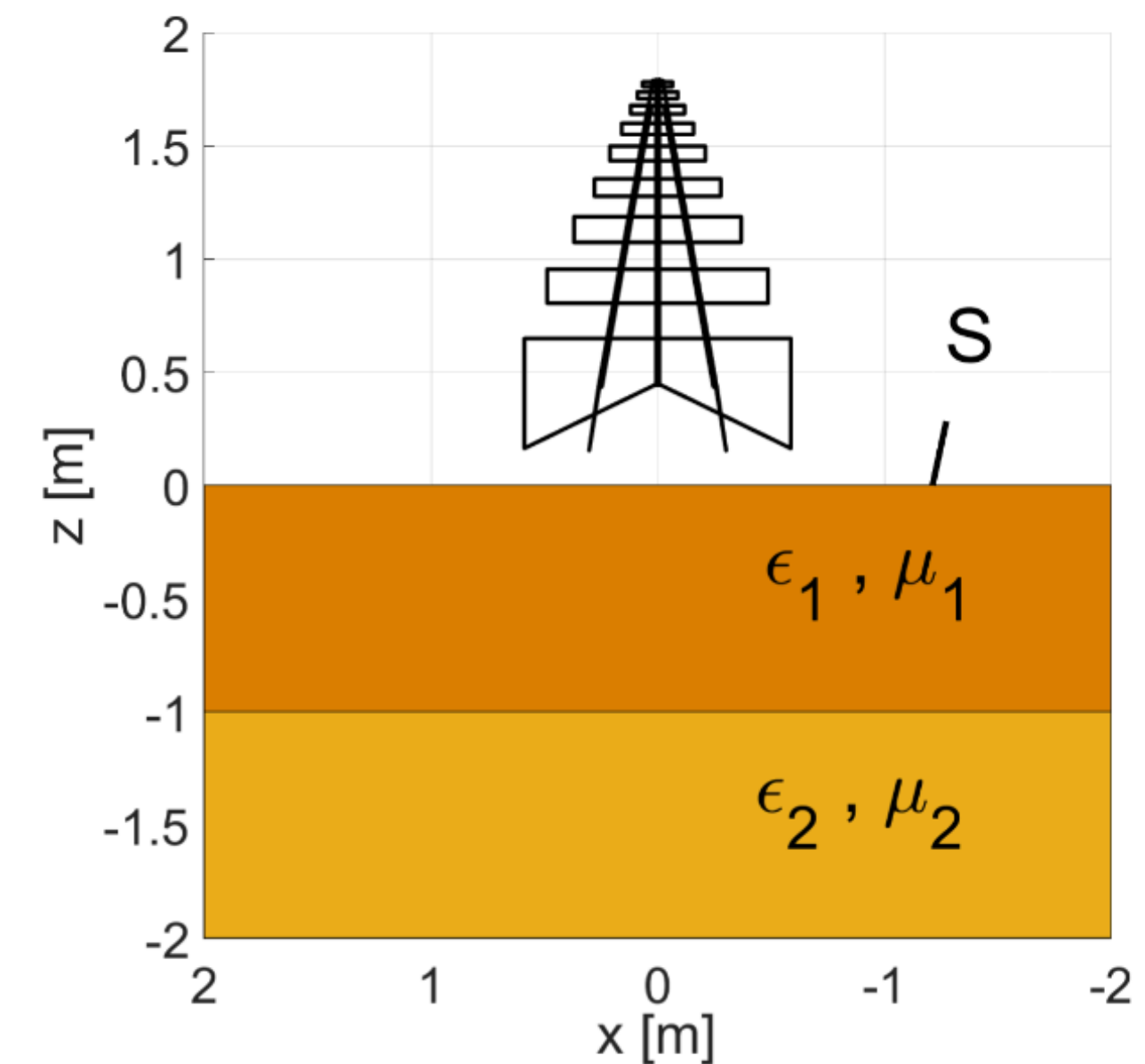
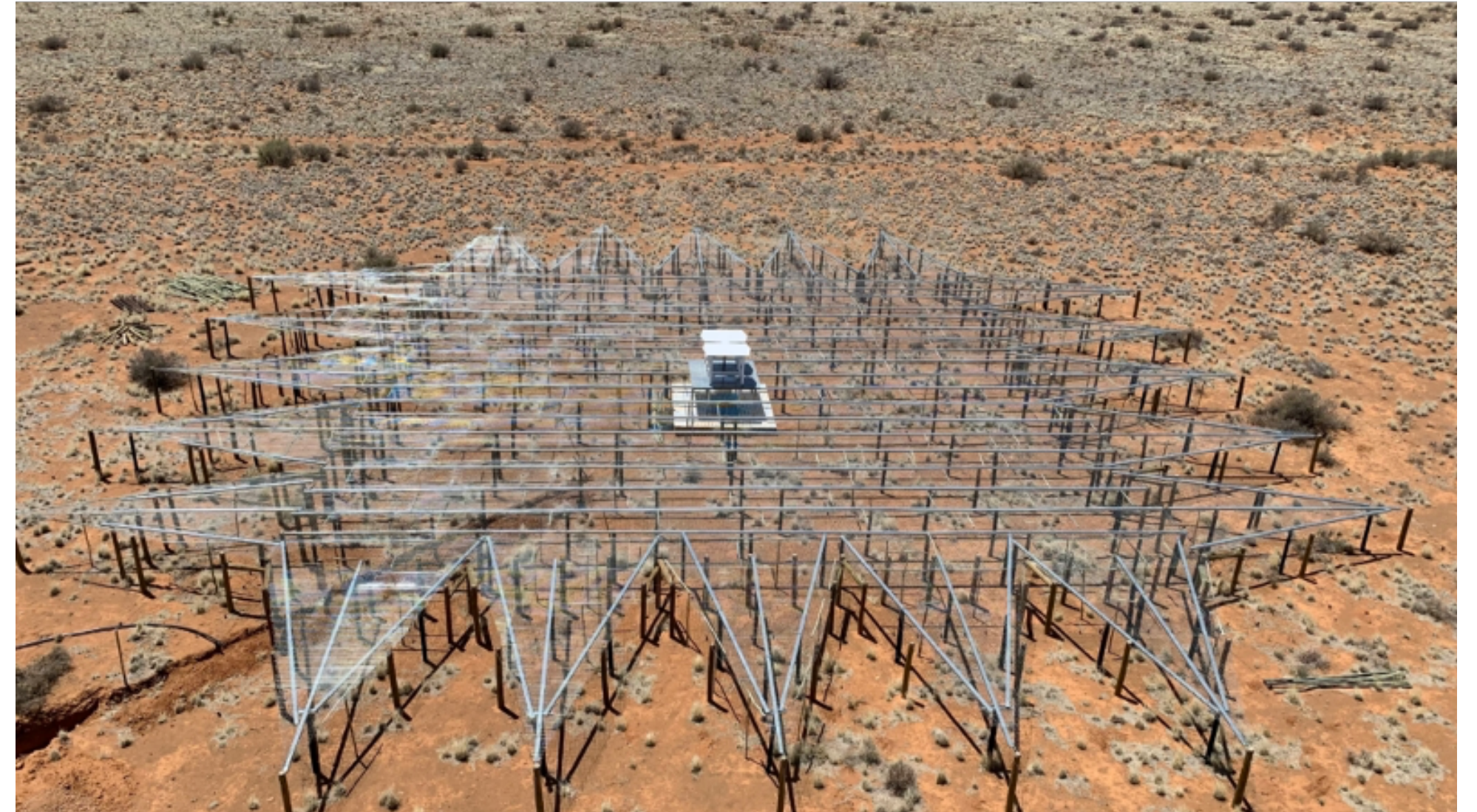


Pattison + Cumner et al. 2026

Fitting for the beam pattern?

Lets fit for the beam...

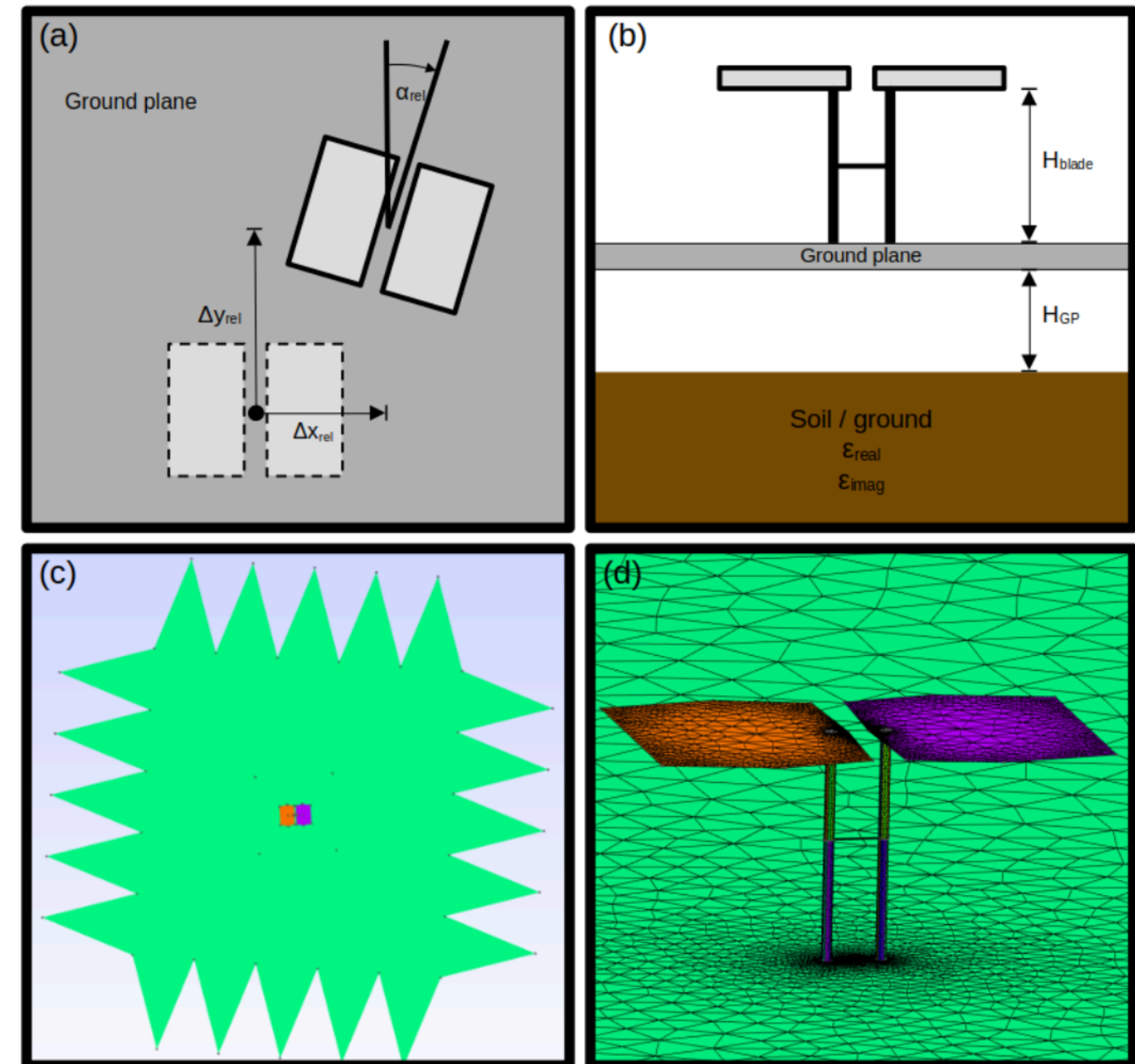
- Ideally we want to fit for
 - the geometry of the antenna
 - And the environment
- CST is very expensive to run and hard to include environmental factors
- Using a MoM simulation code from Cavillot, Jean, et al. 2019 and 2024



Cavillot et al. 2024

What parameters are we varying?

- Generate a suite of simulations with varying
 - Height of the ground plane
 - Height of the blades
 - Rotation and shift of the blades
 - Soil permittivity (correlated with rainfall)

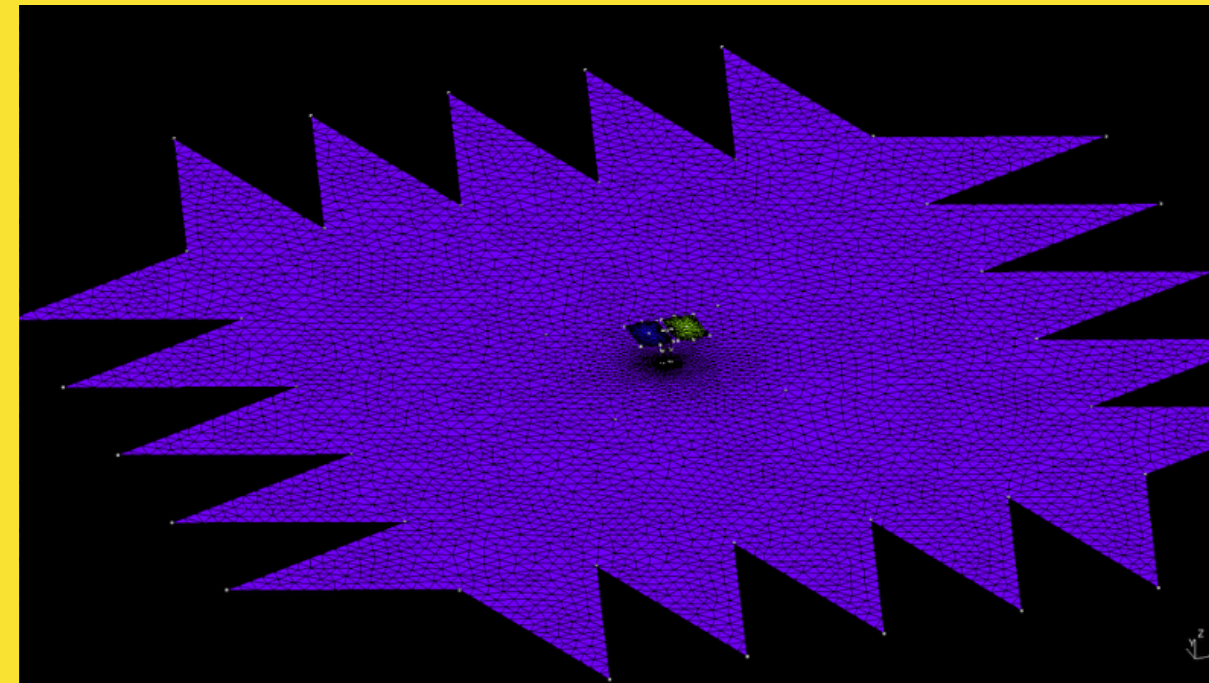


Tutt et al. 2026

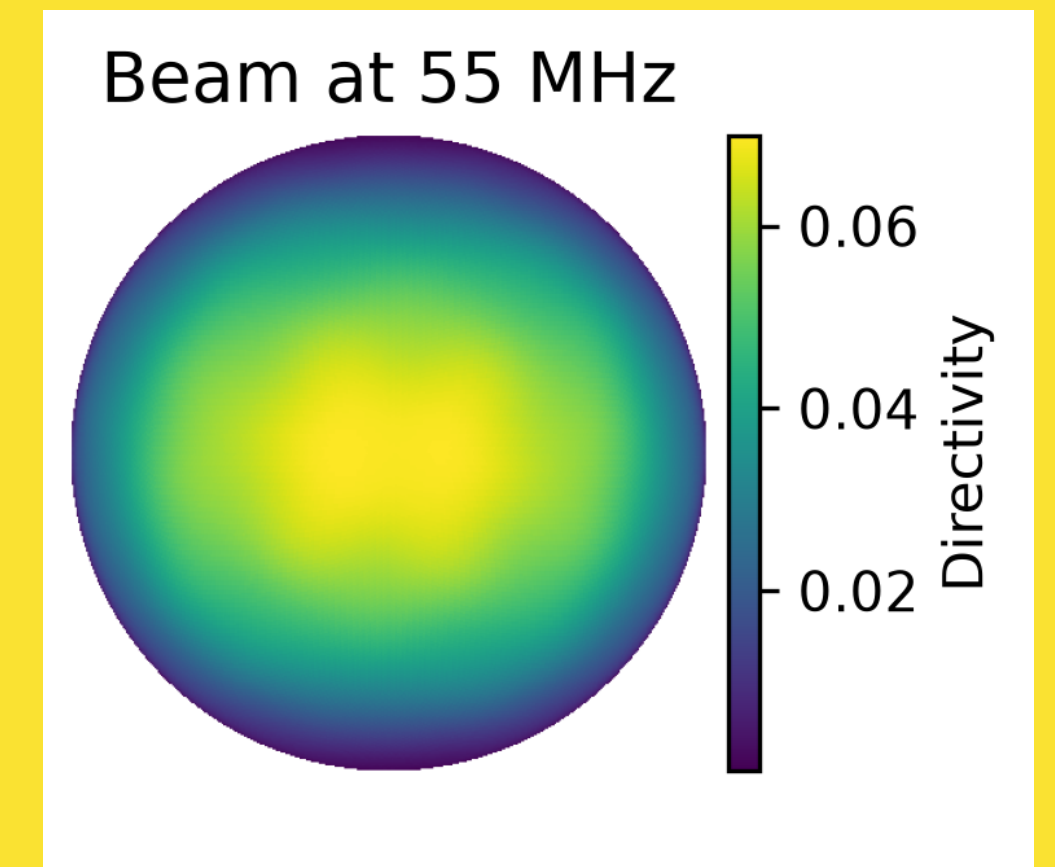
DiRAC RAC17 Allocation - 3.22 million CPU hours



GMSH

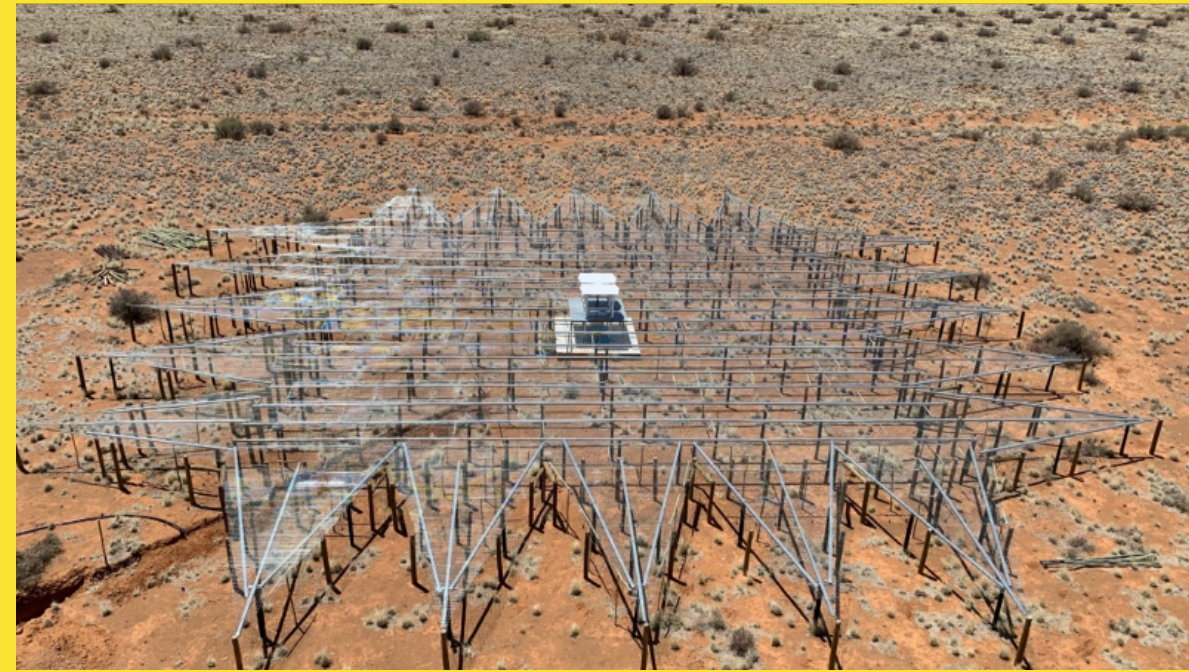


MoM

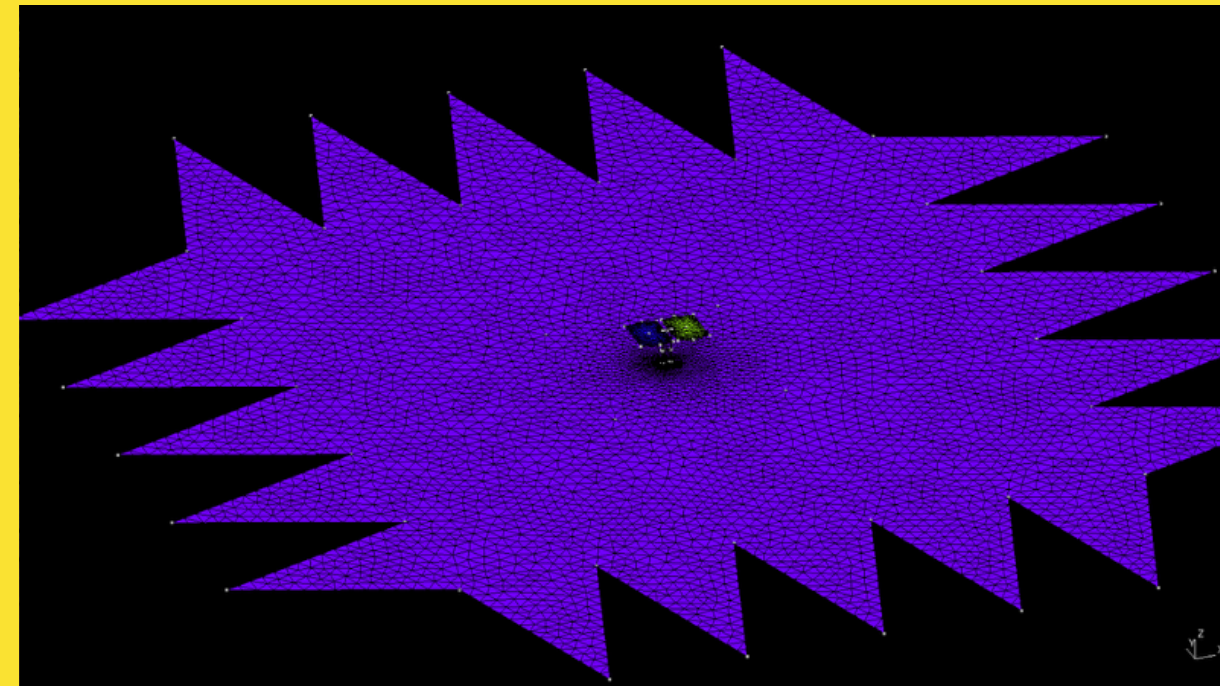


1500 Simulations of REACH-like dipole antennas

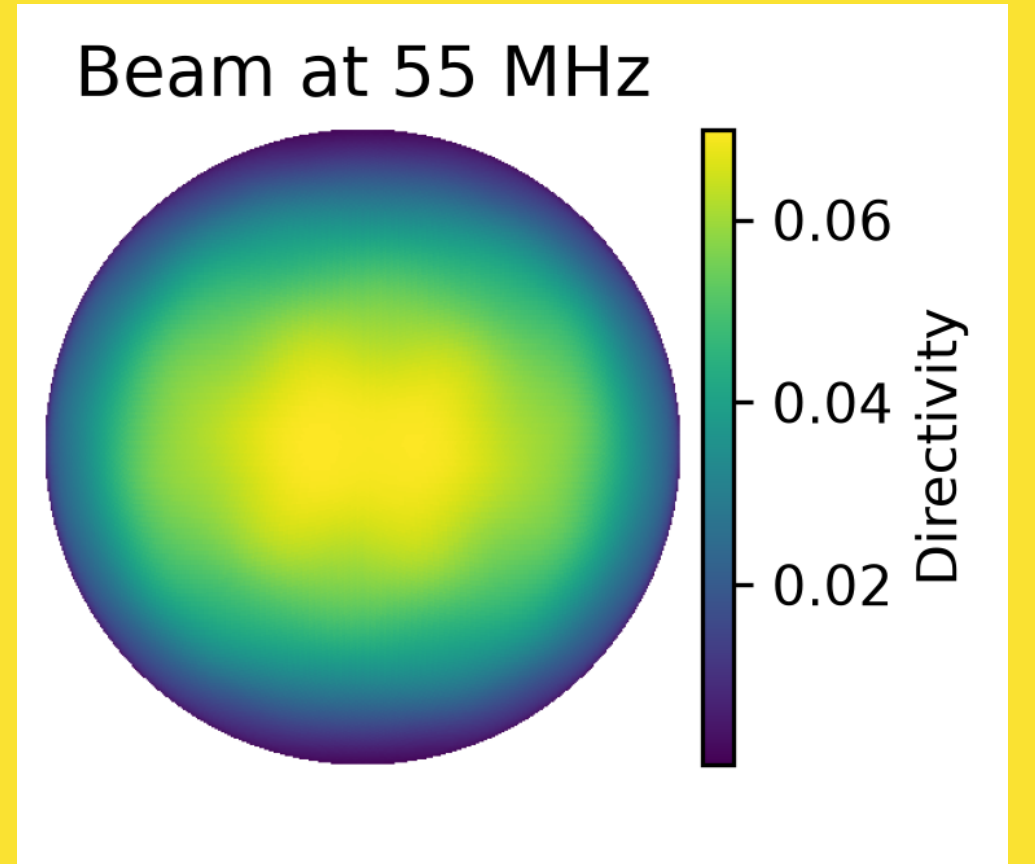
DiRAC RAC17 Allocation - 3.22 million CPU hours



GMSH

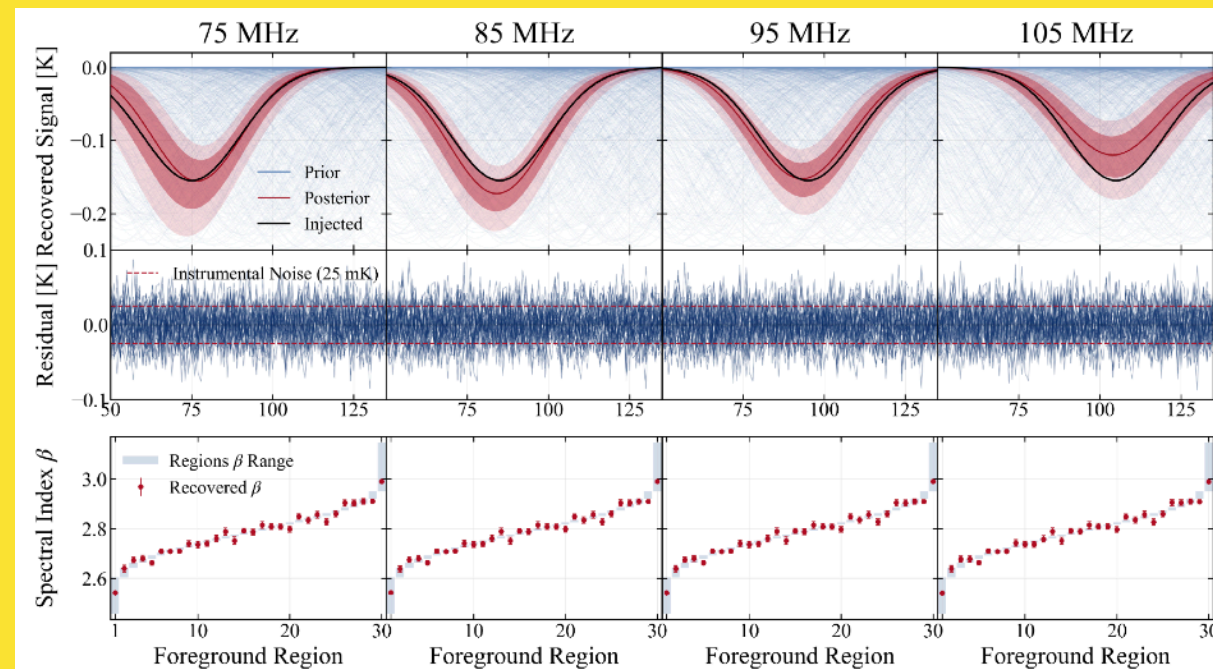


MoM

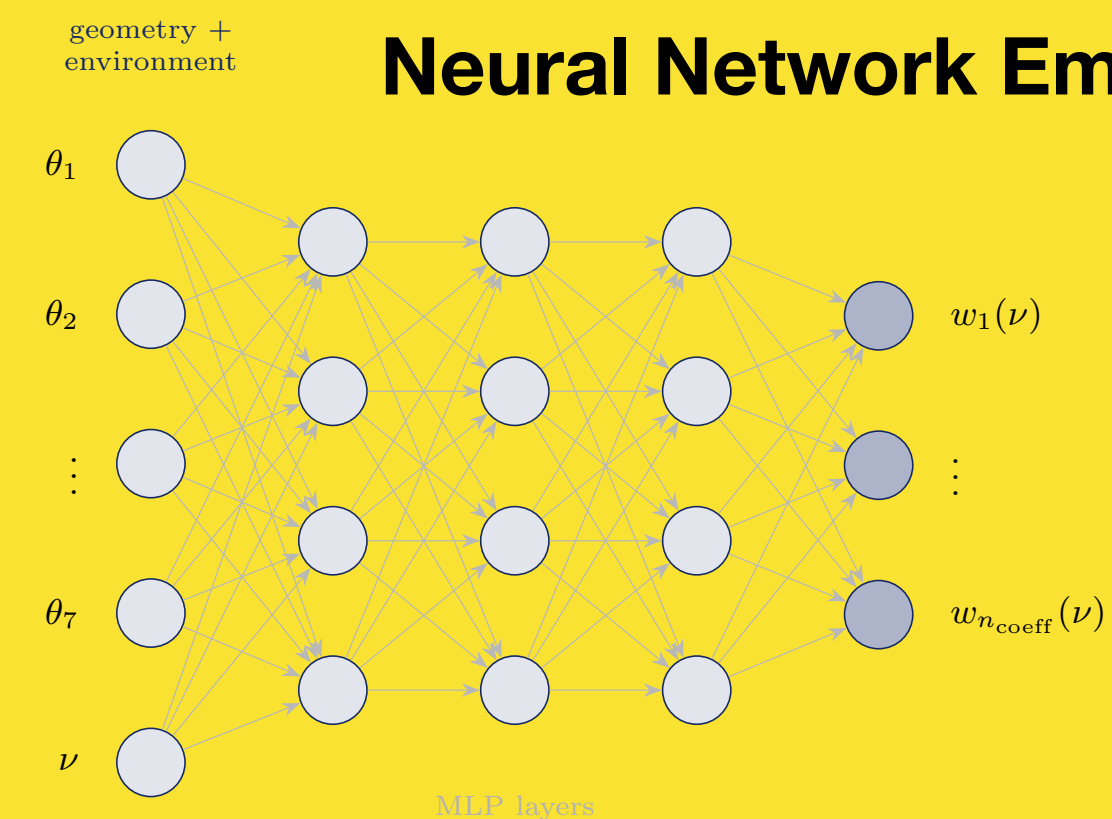


1500 Simulations of REACH-like dipole antennas

Analytic Marginalisation - Jacob's Talk



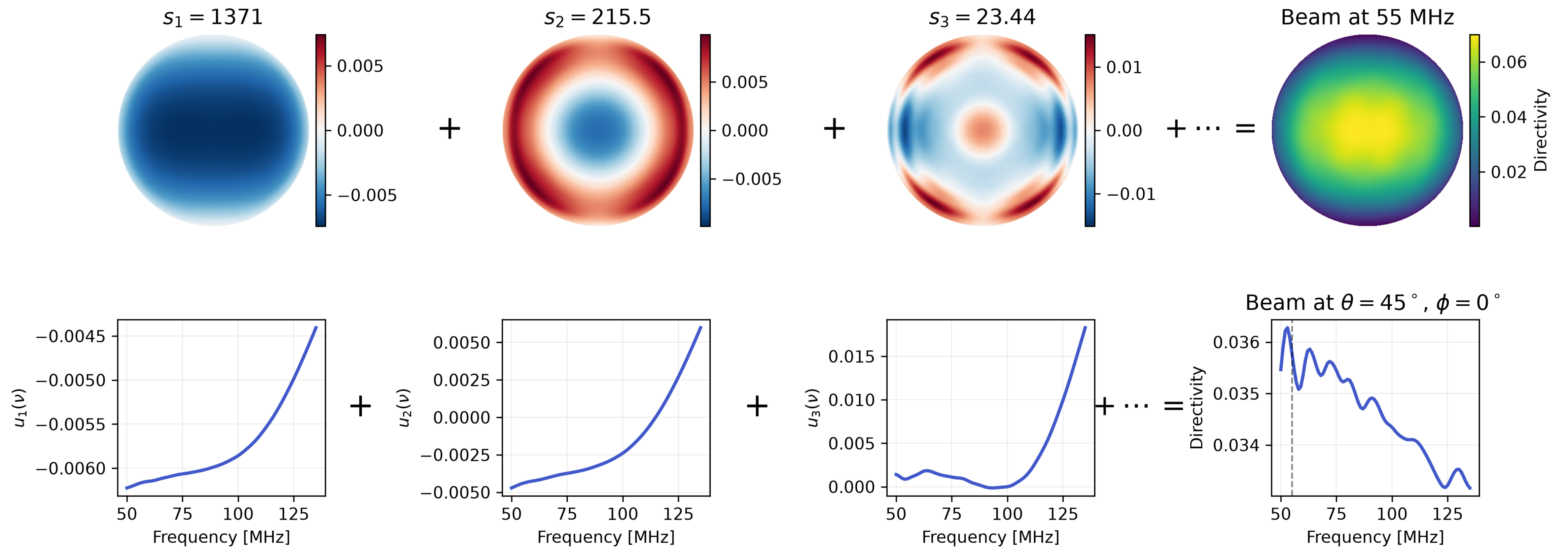
Neural Network Emulation



Designing and training an emulator

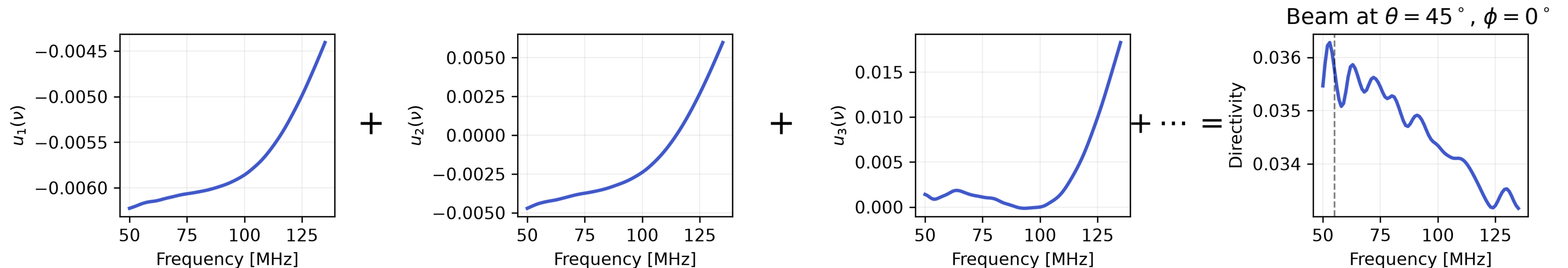
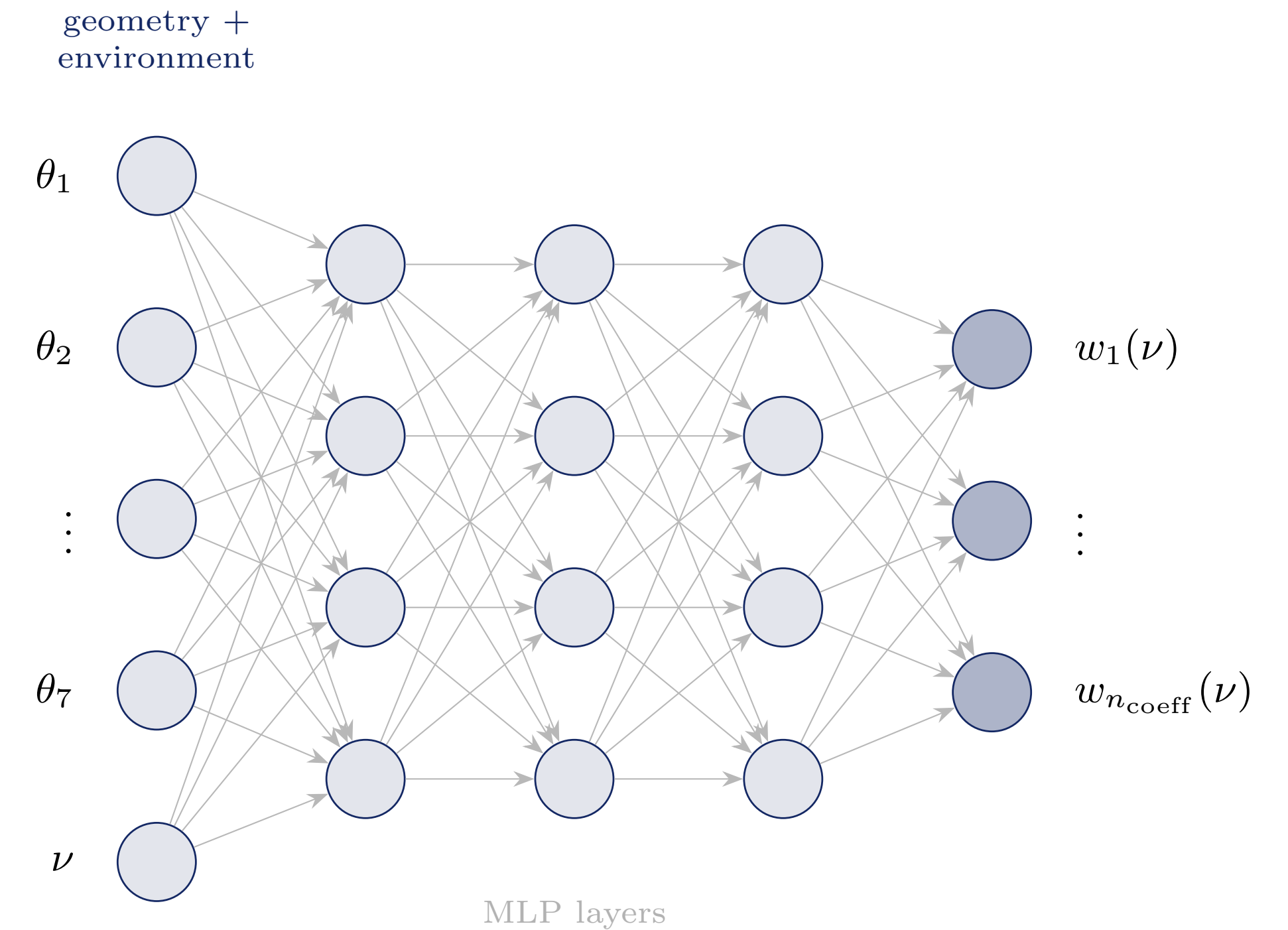
From directivity to SVD...

- Use an SVD across the simulation to derive a set of spatial basis functions and corresponding learnable weights



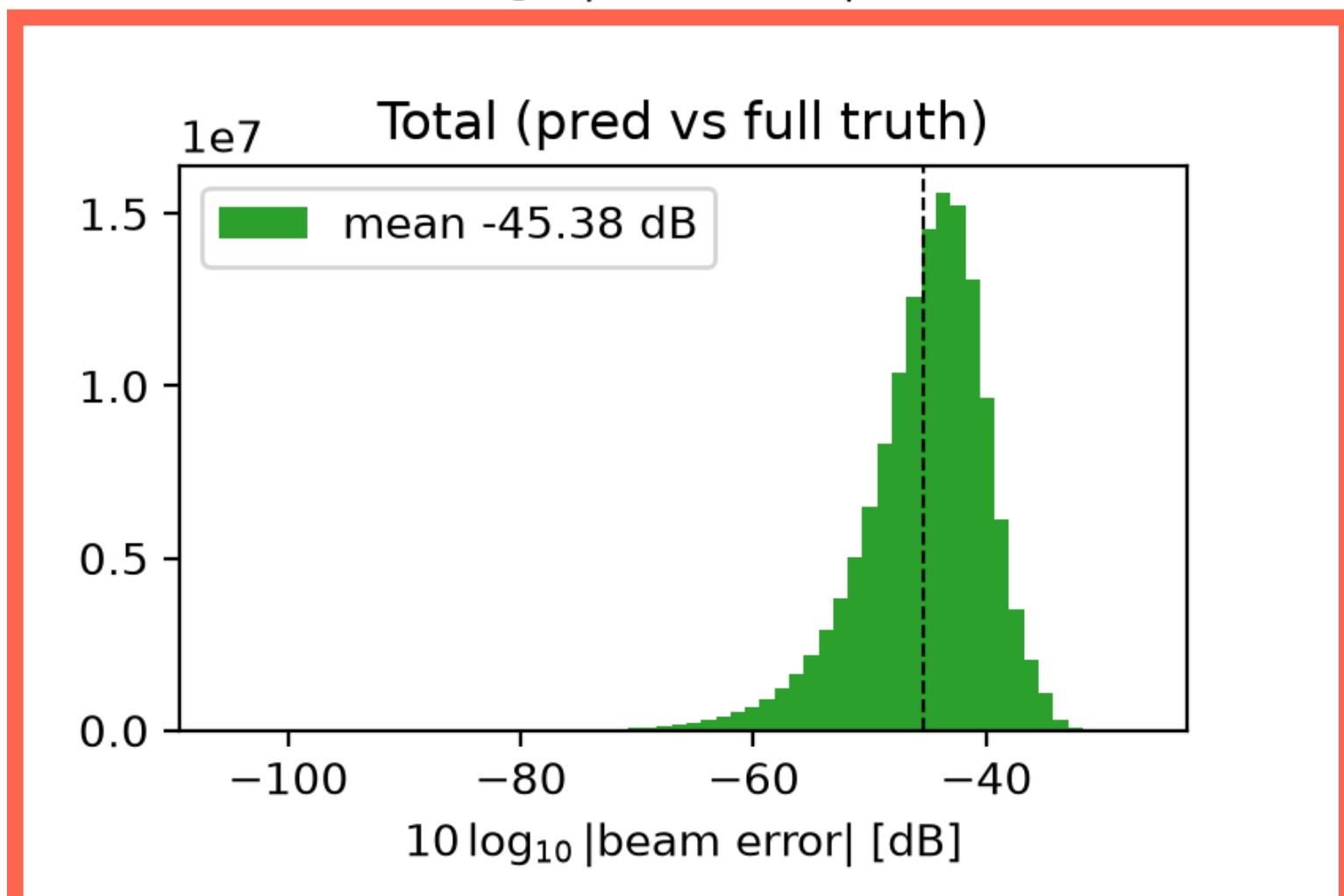
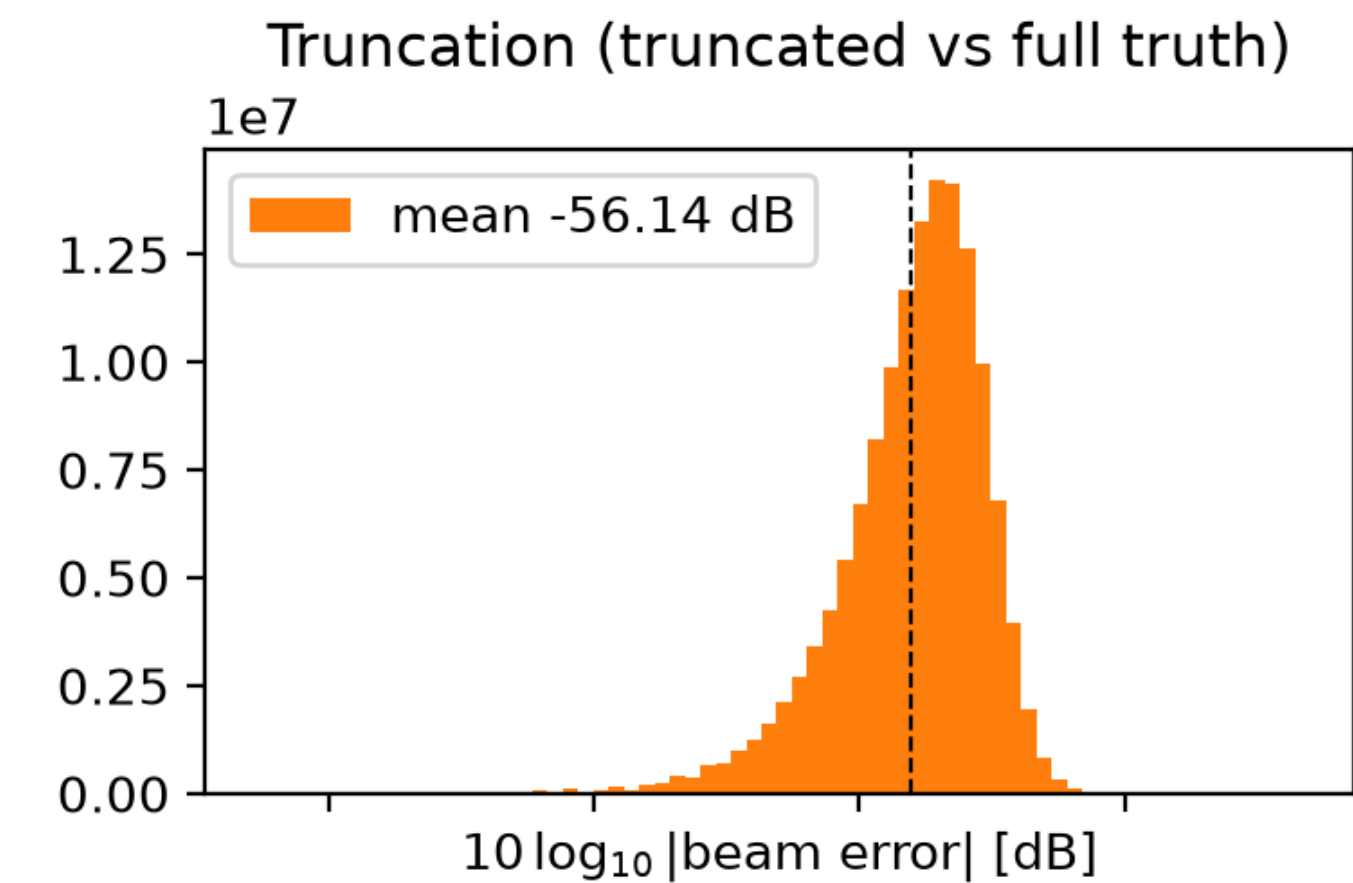
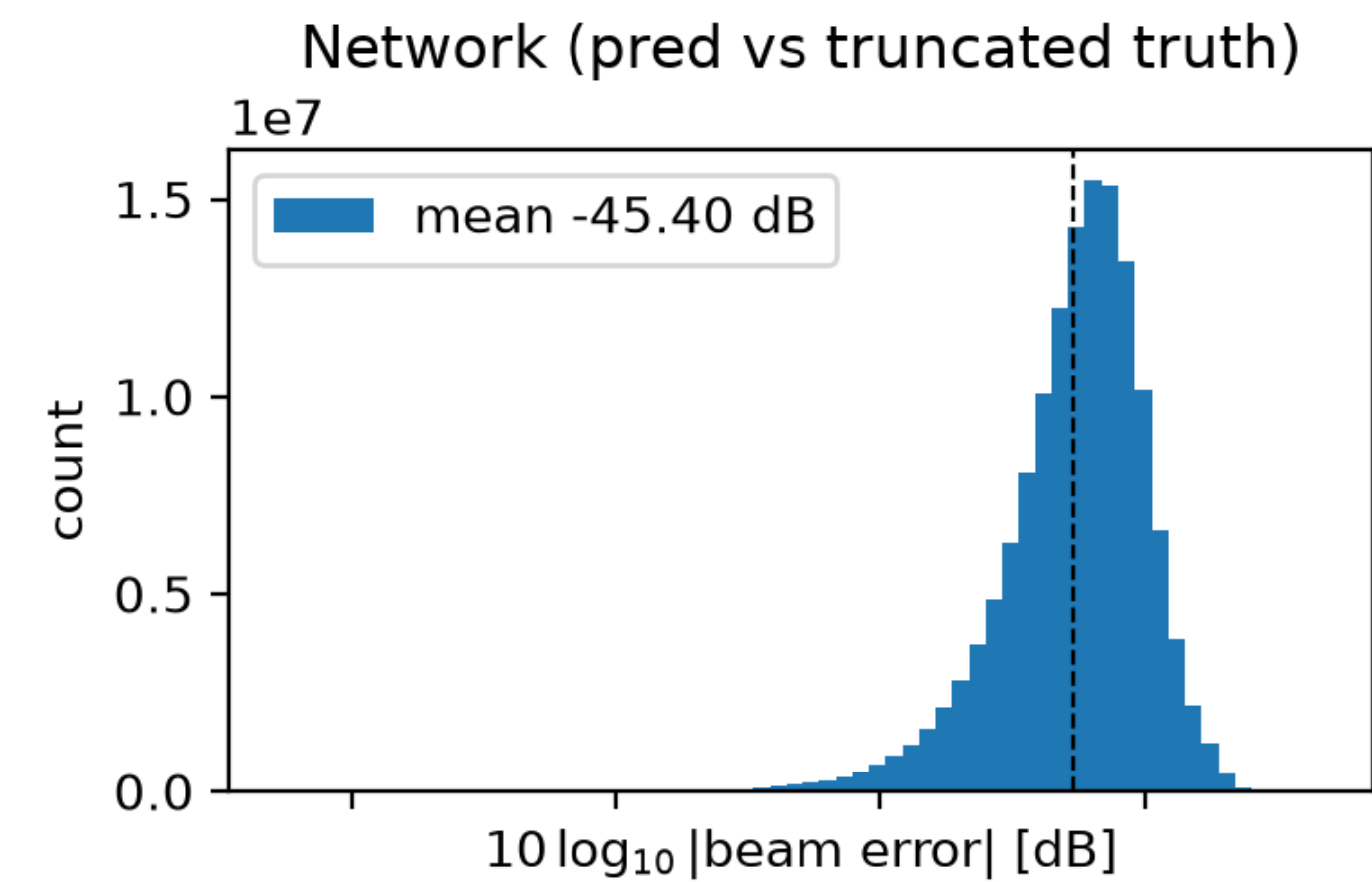
The setup...

- Using an MLP with a tunable architecture
- Takes in geometry + environment parameters + frequency
- Predicts a tunable number of weights $w_i(\nu_j)$ that are combined with the spatial basis functions



Network Performance

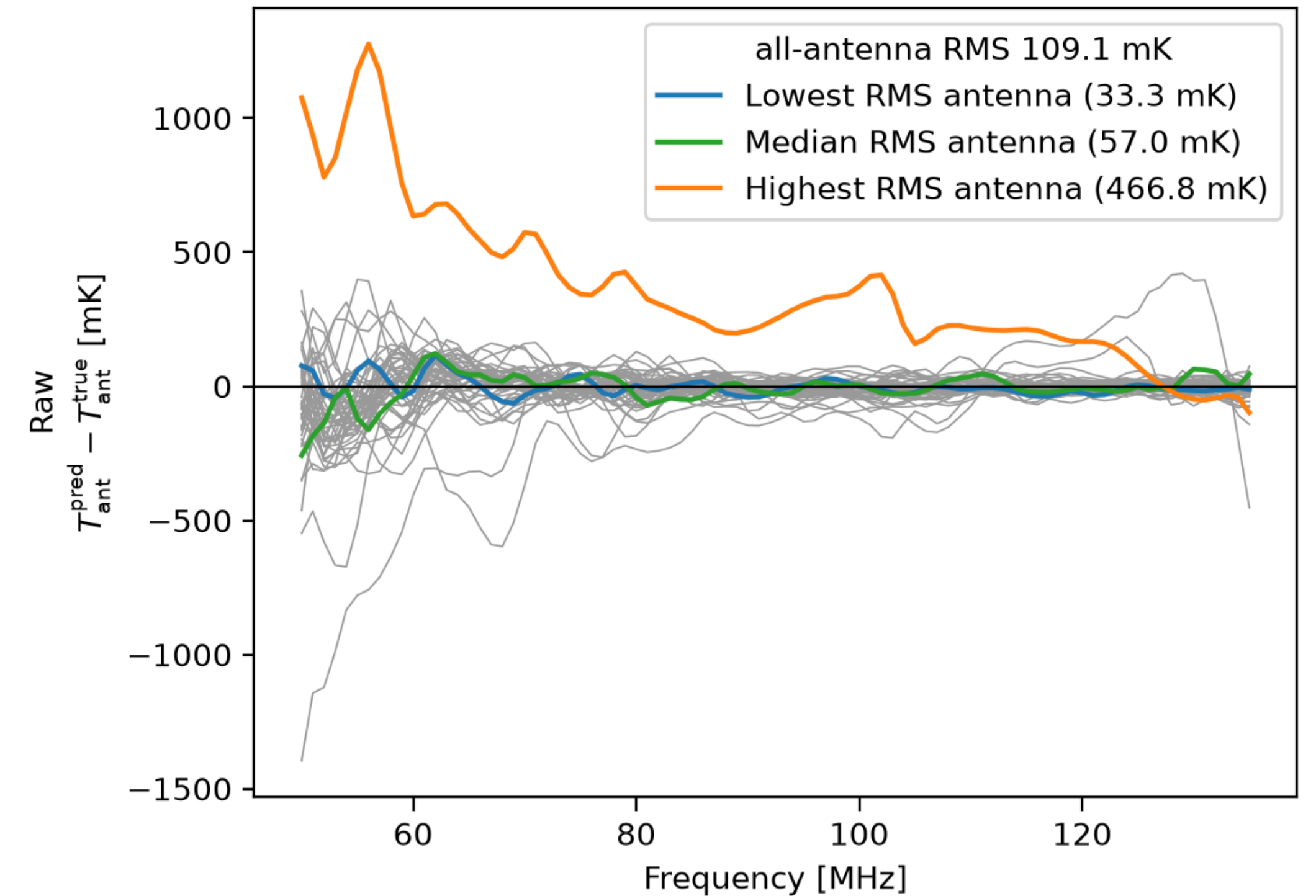
- We train the network on 450 simulations
- Run optuna over the hyper parameter space
- Spilt the simulations into 400 training and 50 validation examples for each trial and hold out an additional 50 for testing
- We can emulator the beam to an accuracy of -45 dB on average



Network Performance

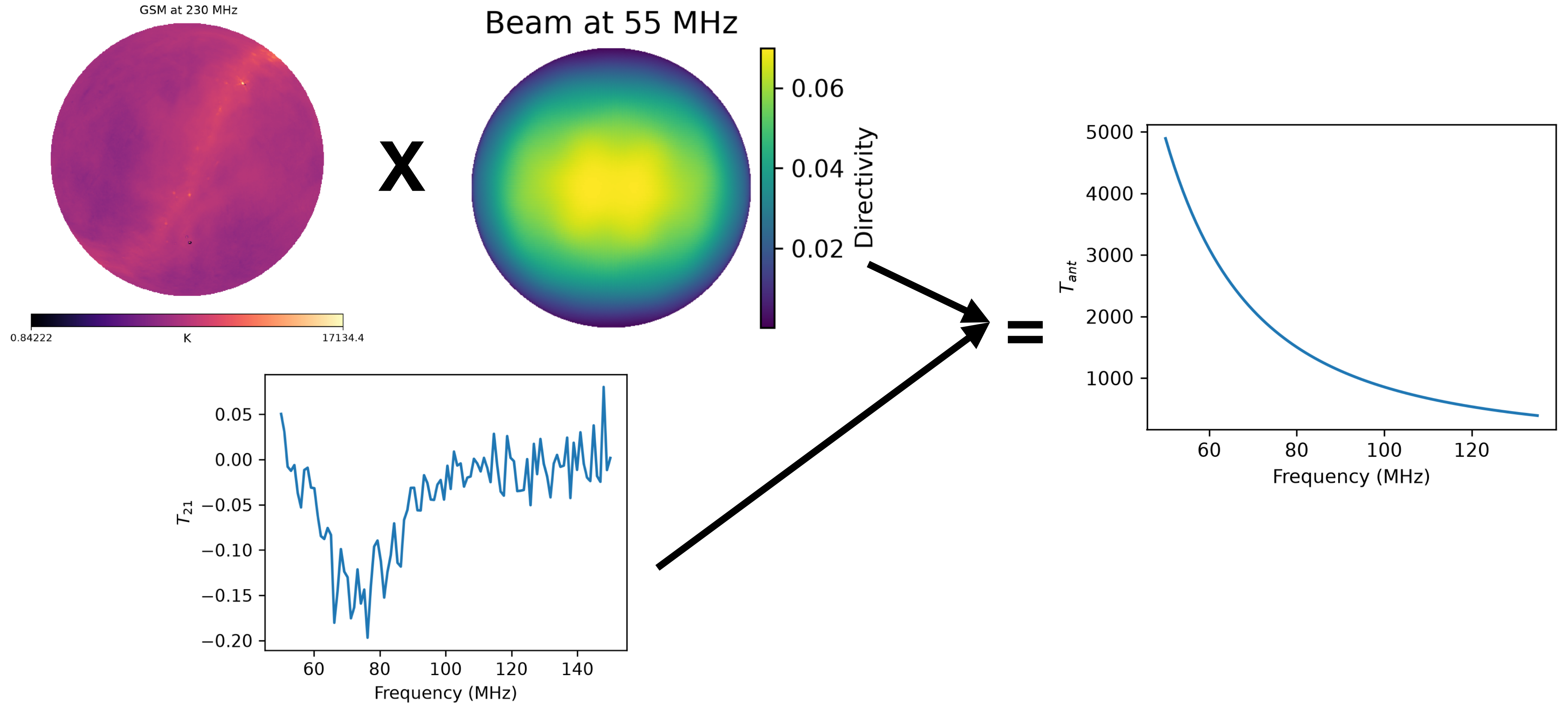
$$T_{\text{ant}}(\nu) = \int T_{\text{sky}}(\nu, \Omega) D(\nu, \Omega) d\Omega$$

- Substitute in $D^{\text{true}}(\nu, \Omega)$ and $D^{\text{pred}}(\nu, \Omega)$
- Take the difference to get an antenna temperature error
- Current network has a median error of 57 mK
- Simplistic uniformly scaled GSM sky



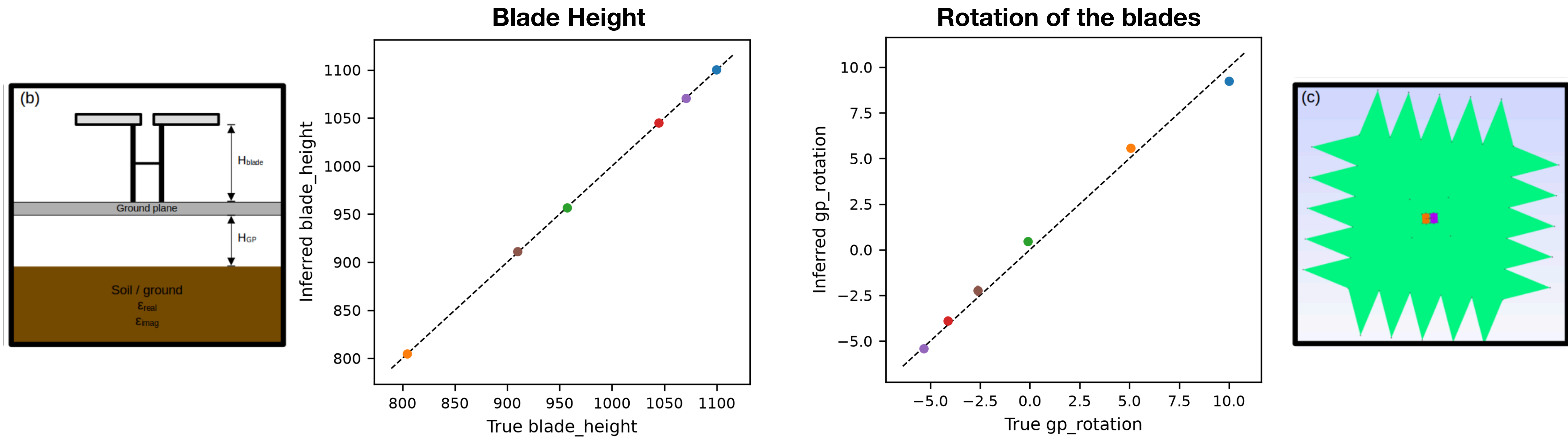
Mock Inference

Mock Inference

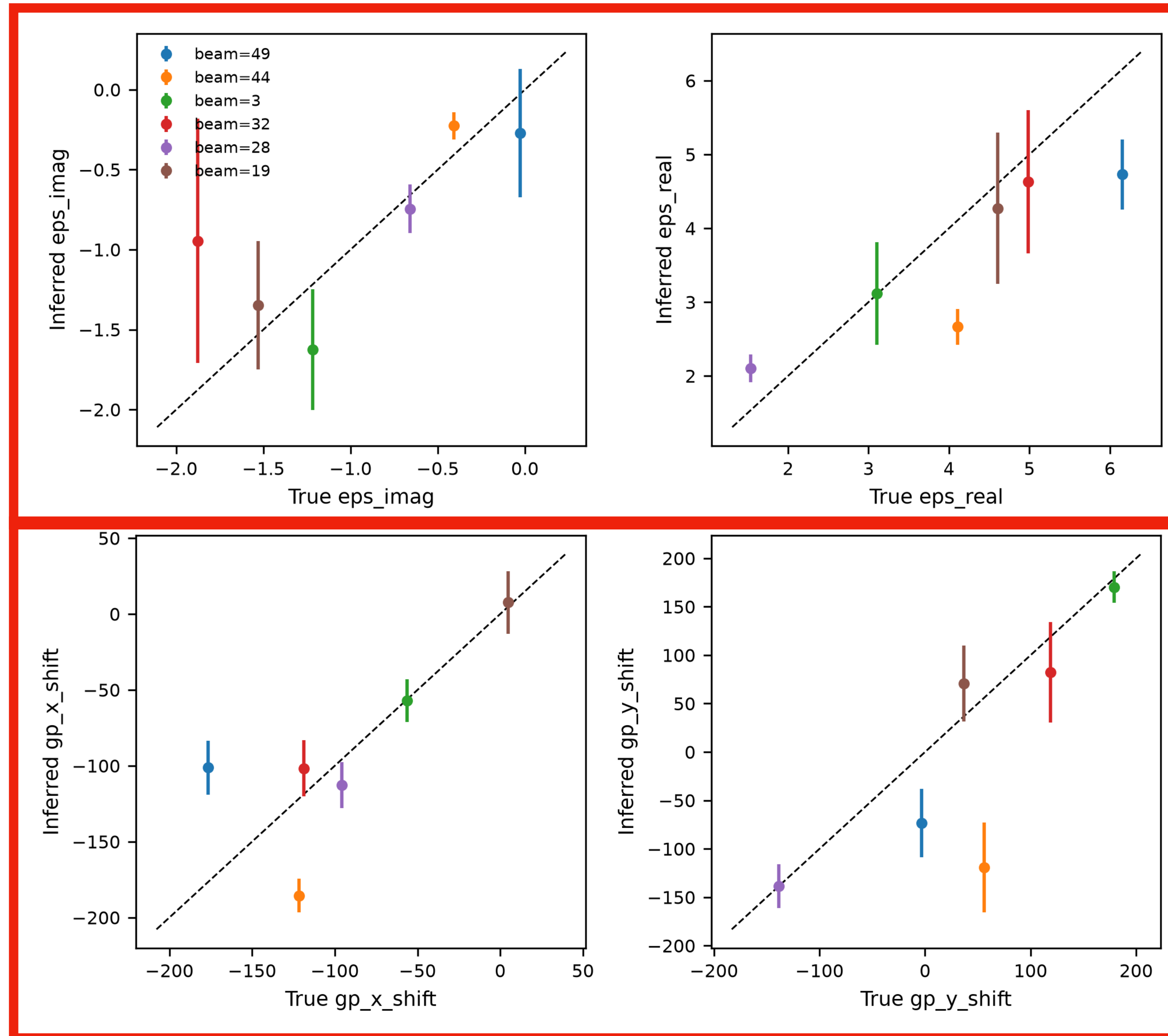


Mock Inference

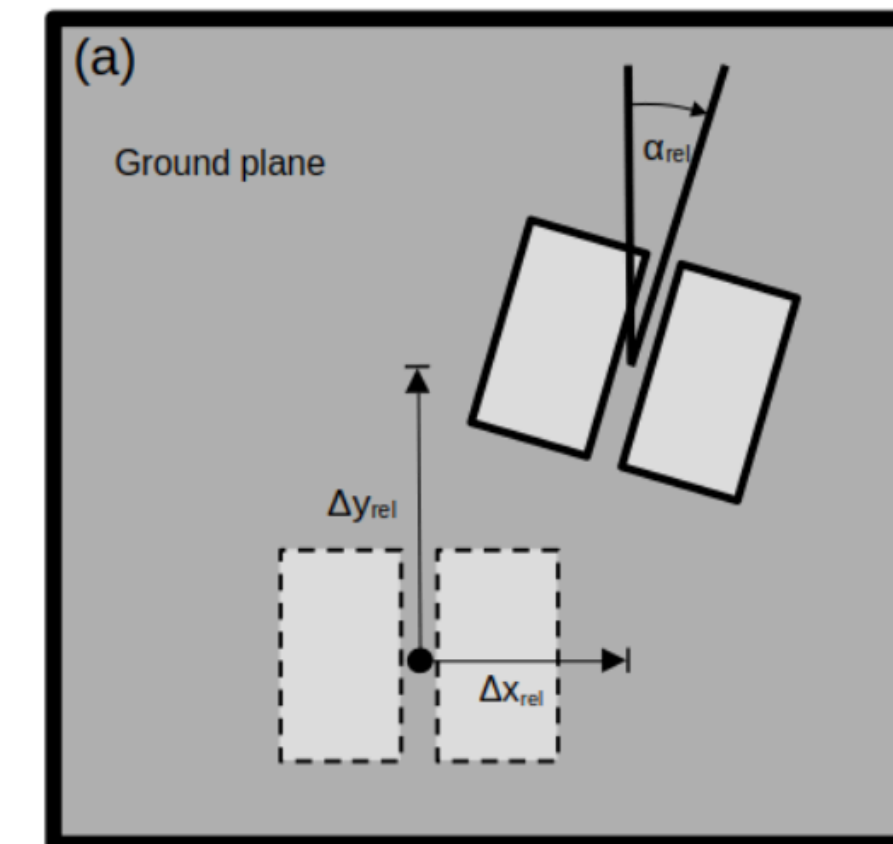
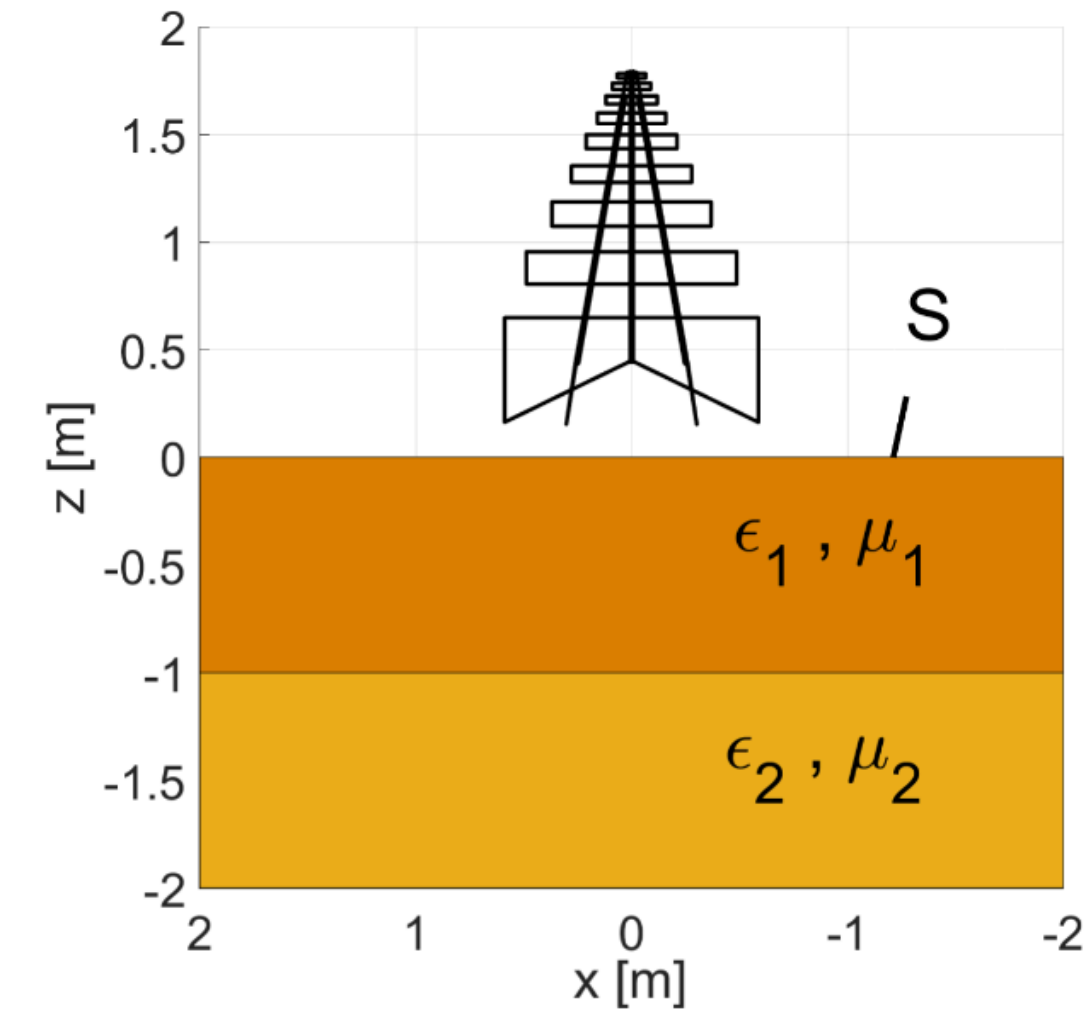
- Replace the beam model with the emulator and run inference using nested sampling
- Tested on 6 random beams

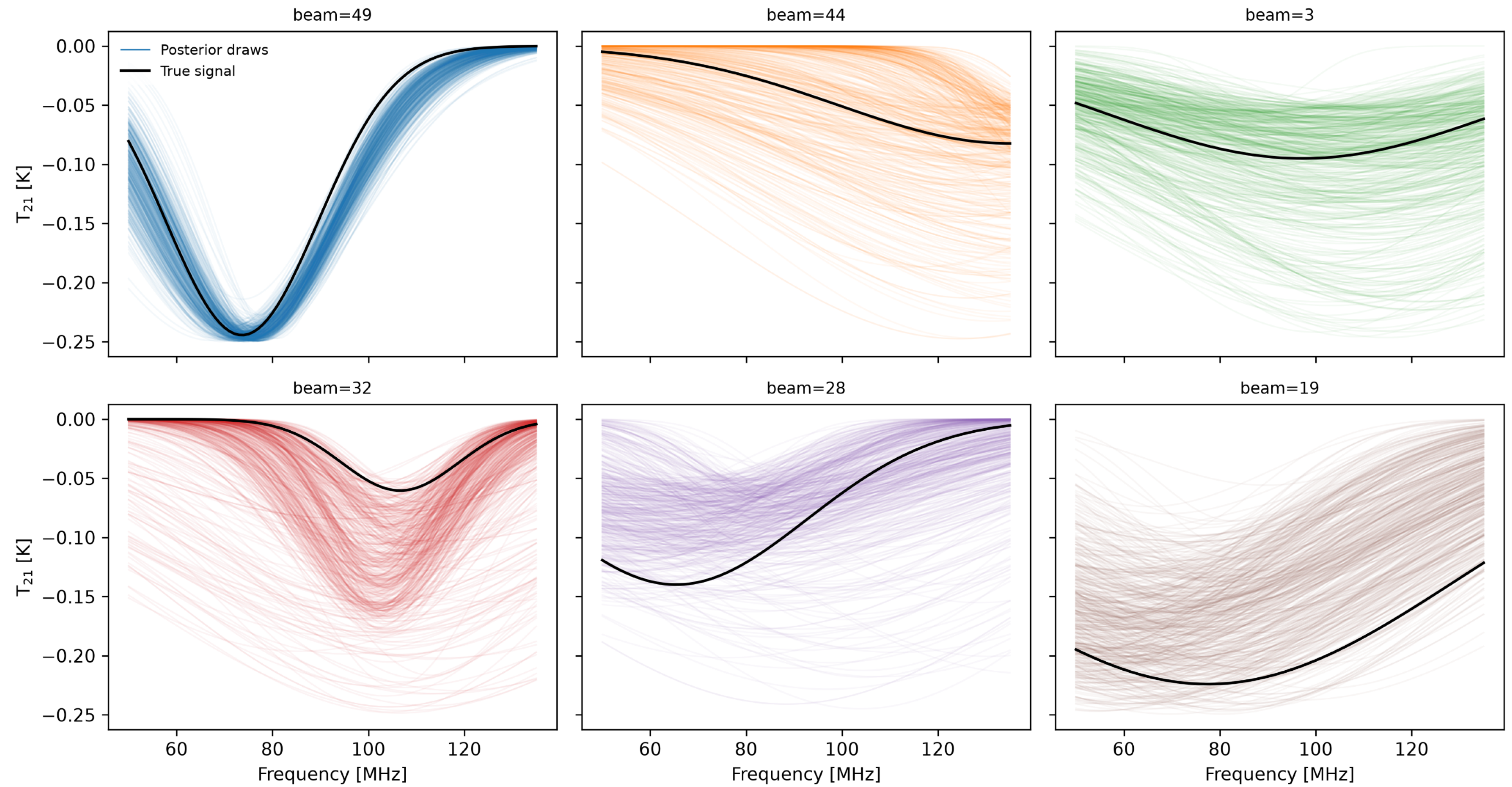


Mock Inference



Properties of the soil





Conclusions

- Paper in preparation and exploring applications to the monopole
- Some room to improve the performance of the emulation
- And testing to do in the REACH pipeline
- But a promising way to handle beam uncertainty
- See Jacob's talk for an alternative way to account for the beams uncertainty

